

```

In [5]: import numpy as np
        from scipy.integrate import odeint
        import matplotlib.pyplot as plt

        def lorenz_system(state, t, sigma, rho, beta):
            x, y, z = state
            dxdt = sigma * (y - x)
            dydt = x * (rho - z) - y
            dzdt = x * y - beta * z
            return [dxdt, dydt, dzdt]

        initial_states = [[1.0, 0.1, 0.1], [5, 5, 5], [-5, -5, 5], [3.5, 3.5, 8.0], [-3.5,
            [7, 7, 7], [-7, -7, 7]
        ]

        time = np.linspace(0, 50, 10000)
        sigma = 10
        rho = 10
        beta = 8 / 3

        fig = plt.figure(figsize=(10, 8))
        ax = fig.add_subplot(111, projection='3d')

        for state in initial_states:
            sol = odeint(lorenz_system, state, time, args=(sigma, rho, beta))
            x, y, z = sol.T
            ax.plot(x, y, z, label=f'Initial = {state}')

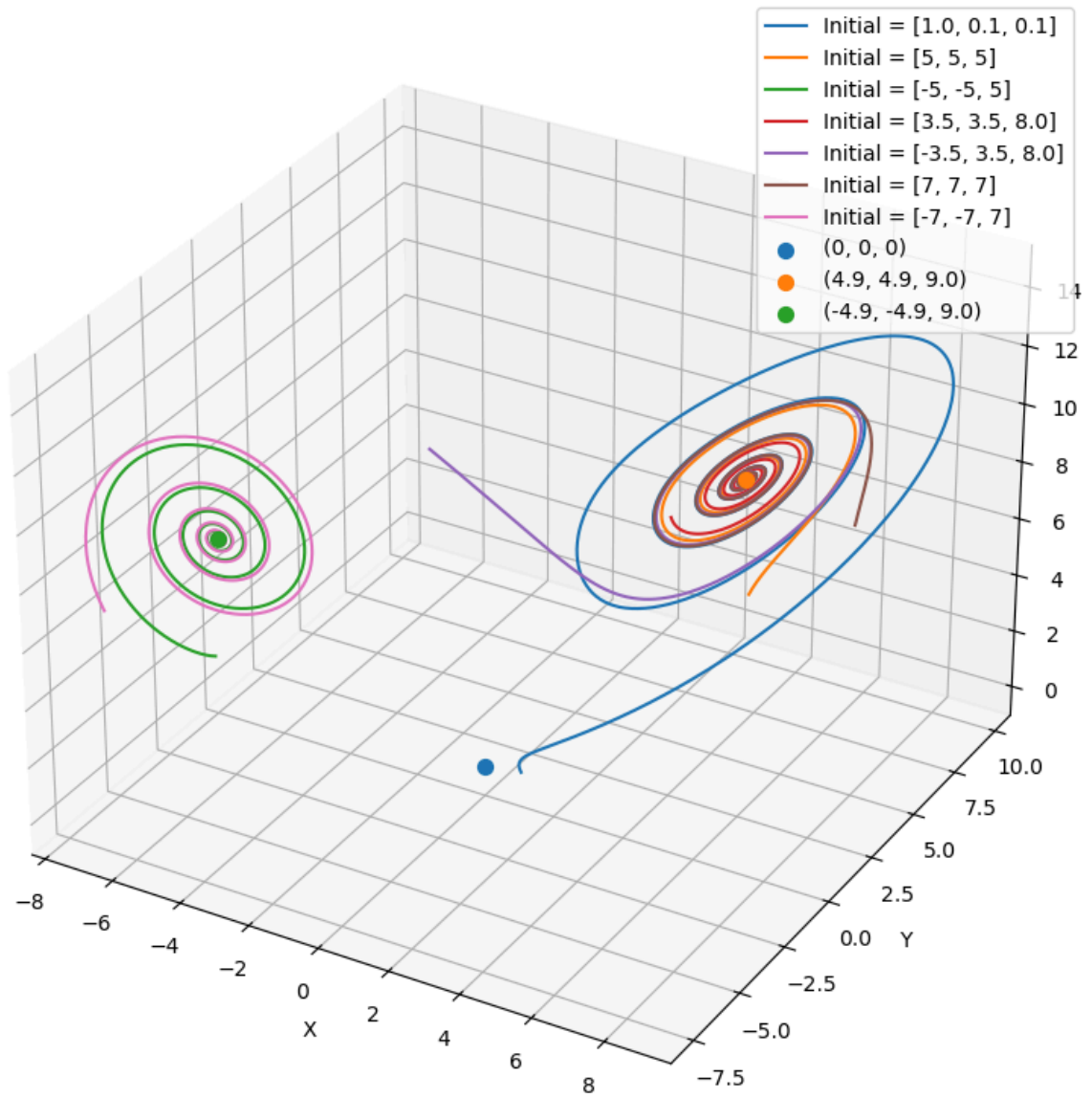
        equil = np.sqrt(beta * (rho - 1))
        equilibria = [[0, 0, 0], [equil, equil, rho - 1], [-equil, -equil, rho - 1]]

        for eq in equilibria:
            ax.scatter(*eq, s=50, label=f'{tuple(np.round(eq, 2))}')

        ax.set_title('Stable points in Lorenz system')
        ax.set_xlabel('X')
        ax.set_ylabel('Y')
        ax.set_zlabel('Z')
        ax.legend()
        plt.tight_layout()
        plt.show()

```

Stable points in Lorenz system



In []: